# Laravel Project

1.Laravel Installation(https://laravel.com/docs/11.x/installation)

- composer global require laravel/installer
- laravel new example-app
- cd example-app
- npm install
- npm run build
- composer run dev

2.Laravel Breeze(https://laravel.com/docs/11.x/starter-kits)

- composer require laravel/breeze --dev
- php artisan breeze:install
- php artisan migrate
- npm install
- npm run dev

3.Laravel adminlte(https://github.com/jeroennoten/Laravel-AdminLTE/wiki/Installation)

- composer require jeroennoten/laravel-adminlte
- php artisan adminlte:install
- composer require laravel/ui
- php artisan ui bootstrap --auth
- php artisan adminlte:install --only=auth_views
- Usage(https://github.com/jeroennoten/Laravel-AdminLTE/wiki/Usage)
    - Paste home.blade.php

```
@extends('adminlte::page')
@section('title', 'Dashboard')
@section('content_header')
    <h1>Dashboard</h1>
@stop
@section('content')
    <p>Welcome to this beautiful admin panel.</p>
@stop
@section('css')
{{-- Add here extra stylesheets --}}
```

Samadhi Perera

{{-- <link rel="stylesheet" href="/css/admin_custom.css"> --}}
@stop
@section('js')
<script> console.log("Hi, I'm using the Laravel-AdminLTE package!"); </script>
@stop

- edit config/adminlte.php
  - Change titile
  - Change logo
  - Change layout_fixed_sidebar =>true
    layout_fixed_Navbar=> true
    layout_fixed_Footer =>true
  - Sidebar_collape =>true
    Sidebar_collape_auto_size =>true
    Sidebar_collape_remember_no_transition=>true
  - Use_route_url=> false
    Dashboard_url =>home
  - Menu=>
    //Navbar item
    Sidebar item
    //blog

```
                    ['text' => 'Dashboard',
                    'url' => 'home',
                    'icon' => 'fas fa-th text-yellow',
        ],
```

```
        [
            'text' => 'User Management',
            'icon' => 'fa fa fa- fa-users text-teal',
            'submenu' => [
                 [
                   'text' => 'Users',
                   'url' => 'users',
                ],

            ],
        ],

        [
            'text' => 'Master Data',
            'icon' => 'fa fa fa- fa-cog',
            'submenu' => [
                [
                    'text' => 'Rank',
                    'url' => 'ranks',
                ],
                [
```

Sachi Perera

```
                'text' => 'Location',
                'url' => 'locations',

            ],
            [
                'text' => 'Unit',
                'url' => 'units',
            ],

        ],
    ],
```

4. Install Spatie(https://spatie.be/docs/laravel-permission/v6/installation-laravel)

- composer require spatie/laravel-permission
- php artisan vendor:publish --provider="Spatie\Permission\PermissionServiceProvider"

  php artisan optimize:clear
  # or
  php artisan config:clear
- php artisan migrate
- Add the necessary trait to your User model:
  // The User model requires this trait
  use Spatie\Permission\Traits\HasRoles;
  use HasRoles;
- php artisan make:seeder RoleAndPermissionSeeder

```php
<?php

namespace Database\Seeders;


use Illuminate\Database\Console\Seeds\WithoutModelEvents;
use Illuminate\Database\Seeder;
use Spatie\Permission\Models\Role;
use Spatie\Permission\Models\Permission;
use App\Models\User;

class RoleAndPermissionSeeder extends Seeder
{
    /**
     * Run the database seeds.
     */
    public function run(): void
    {
        // Create Permissions
        Permission::create(['name' => 'view_users']);
        Permission::create(['name' => 'create_users']);
        Permission::create(['name' => 'edit_users']);
        Permission::create(['name' => 'delete_users']);
        Permission::create(['name' => 'view_ranks']);
        Permission::create(['name' => 'create_ranks']);
```

dhi Perera

```
        Permission::create(['name' => 'edit_ranks']);
        Permission::create(['name' => 'delete_ranks']);
        Permission::create(['name' => 'view_locations']);
        Permission::create(['name' => 'create_locations']);
        Permission::create(['name' => 'edit_locations']);
        Permission::create(['name' => 'delete_locations']);
        Permission::create(['name' => 'view_units']);
        Permission::create(['name' => 'create_units']);
        Permission::create(['name' => 'edit_units']);
        Permission::create(['name' => 'delete_units']);

        // Create Roles
        $adminRole = Role::create(['name' => 'admin']);
        $officerRole = Role::create(['name' => 'officer']);
        $operatorRole = Role::create(['name' => 'operator']);


        // Assign Permissions to Admin
        $adminRole->givePermissionTo(Permission::all());

        // Assign Permissions to Officer
        $officerRole->givePermissionTo([
            'view_users',
            'create_users',
            'edit_users',
            'view_ranks',
            'create_ranks',
            'edit_ranks',
            'view_locations',
            'create_locations',
            'edit_locations',
            'view_units',
            'create_units',
            'edit_units',
        ]);
// Assign Permissions to Operator
        $operatorRole->givePermissionTo([
            'view_users',
            'view_ranks',
            'view_locations',
            'view_units',
        ]);
        $user = User::find(1); // Find user with ID 1
        $user->assignRole('admin'); // Assign 'admin' role

        $user = User::find(2); // Find user with ID 2
        $user->assignRole('officer'); // Assign 'officer' role

        $user = User::find(3); // Find user with ID 3
        $user->assignRole('operator'); // Assign 'operator' role
                }
}
```

- php artisan db:seed --class=RoleAndPermissionSeeder
- php artisan tinker

```
        $user = User::find(1);
        $user->assignRole('admin');
        $user = User::find(2);
```

```
        $user->assignRole('officer');
        $user = User::find(3);
        $user->assignRole('operator');
```

- Models\User.php

```
public function role()
    {
        return $this->belongsTo(Role::class, 'role_id');
           }
```

- php artisan make:model Role

```
  public function permission(){
        return $this->belongsToMany(Permission::class);
    }
    public function user(){
        return $this->hasMany(User::class);
    }
}
```

- php artisan make:model Permission

```
 public function role(){
        return $this->belongsToMany(Role::class, 'role_id');
    }
```

- php artisan make:controller UserController

```
<?php

namespace App\Http\Controllers;
use App\Models\User;
use App\Models\Location;  // Import Location Model
use App\Models\Role;
use App\Models\Rank;
use App\Models\Unit;
use App\DataTables\UsersDataTable;
use Illuminate\Support\Facades\Hash;
use Illuminate\Http\Request;

class UserController extends Controller
{
    public function __construct()
    {
        $this->middleware('permission:view_users')->only('index', 'show');
        $this->middleware('permission:create_users')->only('create', 'store');

        $this->middleware('permission:edit_users')->only('edit', 'update');
        $this->middleware('permission:delete_users')->only('destroy');
    }
    public function index(UsersDataTable $dataTable)
    {
        return $dataTable->render('users.index');
    }
    public function create()
    {
        $location = Location::all();
        $role = Role::all();
        $rank = Rank::all();
        $unit = Unit::all();
```

```
        return view('users.create', compact('location','role','rank','unit'));
     }
    public function store(Request $request)
    {
        $validated = $request->validate([
        'service_no' => 'required|string|max:255',
        'name' => 'required|string|max:255',
        'email' => 'required|email|unique:users,email',
        'password' => 'required|string|min:8',
        'location_id' => ['required', 'numeric'],
        'rank_id' => ['required', 'numeric'],
        'unit_id' => ['required', 'numeric'],
        'password' => ['required', 'string', 'min:8'], // Optional field
        'role_id' => ['required'], // Validate role input
        ]);
        $user = User::create($validated);
        $user->password = Hash::make($request->password);
        $user->active = 1;
        $user->save();

       $role = Role::findOrFail($request->input('role_id'));

        // Sync the roles for the user (in case role is updated)

        $user->syncRoles([$role->name]);
        $user->assignRole($role->name);
        return redirect()->route('users.index');
    }
    public function show(User $user)
    {
        return view('users.show', compact('user'));
    }
    public function edit(User $user)
    {
        $location = Location::all();
        $role = Role::all();
        $rank = Rank::all();
        $unit = Unit::all();
        return view('users.edit', compact('user','location','role','rank','unit'));
    }
    public function update(Request $request, User $user)
    {
        $user_detail = $request->validate([
            'service_no' => ['required', 'string'],
            'name' => 'required|string|max:255',
            'email' => 'required|string|max:255',
            'location_id' => ['required', 'numeric'],
            'rank_id' => ['required', 'numeric'],
            'unit_id' => ['required', 'numeric'],
            'password' => 'required|string|min:8', // Optional field
            'role_id' => ['required'], // Validate role input
        ]);
        $user->update($user_detail);
        $role = Role::findOrFail($request->input('role_id'));
        // Sync the roles for the user (in case role is updated)
        $user->syncRoles([$role->name]);
        $user->assignRole($role->name);

        $user->syncPermissions($request->input('permissions'));
        // $user->update($request->all());
```

```
        return redirect()->route('users.index');
    }
    public function destroy(User $user)
    {
        $user->delete();
        return redirect()->route('users.index');
    }
}
```

- Bootstrap/app.php

```
 ->withMiddleware(function (Middleware $middleware) {
        $middleware->alias([
            'role' => \Spatie\Permission\Middleware\RoleMiddleware::class,
            'permission' => \Spatie\Permission\Middleware\PermissionMiddleware::class,
            'role_or_permission' =>
\Spatie\Permission\Middleware\RoleOrPermissionMiddleware::class,
        ]);
    })
```

- Web.php

```
<?php

use App\Http\Controllers\ProfileController;
use Illuminate\Support\Facades\Route;
use App\Http\Controllers\LocationController;
use App\Http\Controllers\RankController;
use App\Http\Controllers\UnitController;
use App\Http\Controllers\UserController;

Route::get('/', function () {

    return view('auth.login');
});
Route::get('/dashboard', function () {
    return view('dashboard');
})->middleware(['auth', 'verified'])->name('dashboard');

Route::middleware('auth')->group(function () {
    Route::get('/profile', [ProfileController::class, 'edit'])->name('profile.edit');
    Route::patch('/profile', [ProfileController::class, 'update'])->name('profile.update');
    Route::delete('/profile', [ProfileController::class, 'destroy'])->name('profile.destroy');
});
Route::middleware('auth')->group(function () {
Route::resource('locations', LocationController::class);
Route::resource('ranks', RankController::class);
Route::resource('units', UnitController::class);
Route::resource('users', UserController::class);
});
require __DIR__.'/auth.php';

Auth::routes();

Route::get('/home', [App\Http\Controllers\HomeController::class, 'index'])->name('home');
```

5.create migration

- Php artisan make:migration create_user_table
- Edit migration

```php
<?php

use Illuminate\Database\Migrations\Migration;
use Illuminate\Database\Schema\Blueprint;
use Illuminate\Support\Facades\Schema;

class CreateUsersTable extends Migration
{
    public function up()
    {
        Schema::create('users', function (Blueprint $table) {
            $table->id();
            $table->string('service_no');
            $table->string('name');
            $table->string('email')->unique();
            $table->timestamp('email_verified_at')->nullable();
            $table->string('password');
            $table->rememberToken();
            $table->tinyInteger('active')->default(0);
            $table->dateTime('active_date_time')->nullable();
            $table->unsignedBigInteger('role_id')->nullable(); // Use appropriate type
            $table->foreign('role_id')->references('id')->on('roles')->onDelete('cascade');
            $table->unsignedBigInteger('location_id')->nullable(); // Use appropriate type
            $table->foreign('location_id')->references('id')->on('locations')-
>onDelete('cascade'); // Define the action on delete
            $table->unsignedBigInteger('rank_id')->nullable(); // Use appropriate type
            $table->foreign('rank_id')->references('id')->on('ranks')-
>onDelete('cascade');
          $table->unsignedBigInteger('unit_id')->nullable(); // Use appropriate type
            $table->foreign('unit_id')->references('id')->on('units')->onDelete('cascade'); //
Define the action on delete
            // Define the action on delete
            $table->timestamps();
        });
    }
    public function down()
    {
        Schema::dropIfExists('users');
    }
}
```

- php artisan migrate

6.crud

- Create.blade.php

```php
@extends('adminlte::page')

@section('content')
<div class="container mt-5">
    <div class="row justify-content-center">
        <div class="col-md-7">
            @if ($errors->any())
    <div class="alert alert-danger">
        <ul>
            @foreach ($errors->all() as $error)
                <li>{{ $error }}</li>
            @endforeach
```

```
        </ul>
    </div>
@endif

            @if (session('status'))
                <div class="alert alert-success">{{ session('status') }}</div>
            @endif
            <div class="card">
                <div class="card-header">{{ __(' Create User') }}</div>
                <div class="card-body">
                    <form action="{{ route('users.store') }}" method="POST">
                        @csrf
                        <div class="mb-3">
                            <label for="">Service No:</label>
                            <input type="text" name="service_no" required class="form-
control"/>
                        </div>
                        <div class="mb-3">
                            <label for="">Name:</label>
                            <input type="text" name="name" required class="form-control"/>
                        </div>
                        <div class="mb-3">
                            <label for="">Email: </label>
                            <input type="text" name="email" required class="form-control"/>
                        </div>
                         <div class="mb-3">
                            <label for="">Rank: </label>
                            <select name="rank_id" id="rank_id" class="form-control" required>
                                @foreach ($rank as $rank)
                                    <option value="{{ $rank->id }}">{{ $rank->name }}</option>
                                @endforeach
                            </select>
                        </div>
                        <div class="mb-3">
                            <label for="">Unit: </label>
                            <select name="unit_id" id="unit_id" class="form-control" required>
                                @foreach ($unit as $unit)
                                    <option value="{{ $unit->id }}">{{ $unit->name }}</option>
                                @endforeach
                            </select>
                        </div>
                        <div class="mb-3">
                            <label for="">Location: </label>
                            <select name="location_id" id="location_id" class="form-control"
required>

                                @foreach ($location as $location)
                                    <option value="{{ $location->id }}">{{ $location->name
}}</option>

                                @endforeach
                            </select>
                        </div>
                        <div class="mb-3">
                            <label for="">Role: </label>
                            <select name="role_id" id="role_id" class="form-control" required>
                                @foreach ($role as $role)
                                    <option value="{{ $role->id }}">{{ $role->name }}</option>
                                @endforeach
                            </select>
                        </div>
                        <div class="mb-3">
```

```
                        <label for="">Password: </label>
                        <input type="text" name="password" required class="form-control"/>
                    </div>

                    <div class="mb-3">
                        <button type="submit" class="btn btn-primary">Save</button>
                    </div>
                </form>


            </div>
        </div>
    </div>
</div>
```

- Edit.blade.php

```
@extends('adminlte::page')

@section('content')
<div class="container mt-5">
    <div class="row justify-content-center">
        <div class="col-md-7">
            @if ($errors->any())
    <div class="alert alert-danger">
        <ul>
            @foreach ($errors->all() as $error)
                <li>{{ $error }}</li>
            @endforeach
        </ul>
    </div>
      @endif
            @if (session('status'))
                <div class="alert alert-success">{{ session('status') }}</div>
            @endif
            <div class="card">
                <div class="card-header">{{ __(' Edit User') }}</div>
                <div class="card-body">
                    <form action="{{ route('users.update', $user->id) }}" method="POST">
                        @csrf
                        @method('PUT')
                        <div class="mb-3">
                            <label for="">Service No:</label>
                            <input type="text" name="service_no" id="service_no" class="form-
control" value="{{ old('service_no', $user->service_no) }}" required>
                        </div>
                        <div class="mb-3">
                            <label for="">Name:</label>
                            <input type="text" name="name" id="name" class="form-control"
value="{{ old('name', $user->name) }}" required>
                        </div>
                        <div class="mb-3">
                            <label for="">Email: </label>
                            <input type="text" name="email" id="email" class="form-control"
value="{{ old('email', $user->email) }}" required>
                            <div class="mb-3">
                                <label for="">Rank: </label>
                                <select name="rank_id" id="rank_id" class="form-control"
required>
```

```
                                    @foreach ($rank as $rank)
                                        <option value="{{ $rank->id }}">{{ $rank->name
}}</option>
                                    @endforeach
                                </select>
                            </div>
                            <div class="mb-3">
                                <label for="">Unit: </label>
                                <select name="unit_id" id="unit_id" class="form-control"
required>
                                    @foreach ($unit as $unit)
                                        <option value="{{ $unit->id }}">{{ $unit->name
}}</option>
                                    @endforeach
                                </select>
                            </div>
                            <div class="mb-3">
                                <label for="">Location: </label>
                                <select name="location_id" id="location_id" class="form-
control" required>
                                    @foreach ($location as $location)
                                        <option value="{{ $location->id }}">{{ $location->name
}}</option>
                                    @endforeach
                                </select>
                            </div>
                            <div class="mb-3">
                                <label for="">Role: </label>
                                <select name="role_id" id="role_id" class="form-control"
required>
                                    @foreach ($role as $role)
                                        <option value="{{ $role->id }}">{{ $role->name
}}</option>
                                    @endforeach
                                </select>
                            </div>

                        </div>
                        <div class="mb-3">
                            <label for="">Password: </label>
                            <input type="text" name="password" id="password" class="form-
control" value="{{ old('password', $user->password) }}"  required>

                        </div>

                        <div class="mb-3">
                            <button type="submit" class="btn btn-primary">Update</button>
                        </div>
                    </form>
                </div>
            </div>
        </div>
    </div>
</div>
```

- Index.blade.php

```
@extends('adminlte::page')
```

```
@section('content')
<div class="container mt-5">
    <div class="row justify-content-center">
        <div class="col-md-12">

            @if (session('status'))
                <div class="alert alert-success">{{ session('status') }}</div>
            @endif
            <div class="card mt-3">
                <div class="card-header">{{ __('Users') }}</div>

                <div class="card-body">
                    <a href="{{ route('users.create') }}" class="btn btn-primary float-end">Add
New User</a>

                    {{-- <table class="table table-bordered table-striped">
                        <thead>
                            <tr>
                                <th>Id</th>
                                <th>Service_no</th>
                                <th>Name</th>
                                <th>Email</th>
                                <th>Active</th>
                                <th>Role</th>
                                <th>Rank</th>
                                <th>Location</th>
                                <th>Unit</th>
                                <th>Created_at</th>
                                <th>Updated_at</th>
                                <th>Actions</th>
                            </tr>
                        </thead>
                        <tbody>
                            @foreach ($users as $user)

                <tr>
                    <td>{{ $user->id }}</td>
                    <td>{{ $user->service_no }}</td>
                    <td>{{ $user->name }}</td>
                    <td>{{ $user->email }}</td>
                    <td>{{ $user->active }}</td>
                    <td>{{ $user->role->name }}</td>
                    <td>{{ $user->ranks->name }}</td>
                    <td>{{ $user->locations->name }}</td>
                    <td>{{ $user->units->name }}</td>
                    <td>{{ $user->created_at }}</td>
                    <td>{{ $user->updated_at }}</td>

                    <td>
                        <a href="{{ route('users.edit', $user->id) }}" class="btn btn-xs btn-
warning" data-toggle="tooltip" title="Edit User" ><i class="fa fa-pen"></i></a>
                                    <form action="{{ route('users.destroy', $user->id) }}"
method="POST" style="display:inline;" onsubmit="return confirmDelete()" >
                                        @csrf
                                        @method('DELETE')
                                        <button type="submit"   class="btn btn-xs btn-danger"
data-toggle="tooltip"  title="Delete User" ><i class="fa fa-trash"></i></button>
                                    </form>
                                    <a href="{{ route('users.show', $user->id) }}" class="btn
btn-xs btn-info" data-toggle="tooltip" title="Edit User" ><i class="fa fa-eye"></i></a>
```

```
                            </td>
                        </tr>
                    @endforeach
                </tbody>
            </table>  --}}

            <div class="table-responsive">
            {{ $dataTable->table() }}

        </div>
            </div>
        </div>
    </div>
    @endsection
    {{-- <script>
        function confirmDelete(){
            return confirm('Are You sure you want to delete this record? This action be
undone.');
        }
    </script> --}}

@push('js')
@section('plugins.Datatables', true)
{{ $dataTable->scripts() }}
@endpush
```

- Show.blade.php

```
@extends('adminlte::page')

@section('content')
<div class="container mt-5">
    <div class="row justify-content-center">
        <div class="col-md-7">

            @if (session('status'))
                <div class="alert alert-success">{{ session('status') }}</div>
            @endif
            <div class="card">
                <div class="card-header"><strong>{{ $user->id }}</strong>
                </div></div>
                <div class="card-body">
                    <ul class="list-group">
                        <li class="list-group-item">
                            <strong>Service No:</strong> {{ $user->service_no }}
                        </li>
                        <li class="list-group-item">
                            <strong>Name:</strong> {{ $user->email }}
                        </li>
                        <li class="list-group-item">
                            <strong>Email:</strong> {{ $user->email }}
                        </li>
                        <li class="list-group-item">
                            <strong>Active:</strong> {{ $user->active }}
                        </li>
                        <li class="list-group-item">
                            <strong>Role:</strong> {{ $user->role->name }}
                        </li>
                        <li class="list-group-item">
                            <strong>Rank:</strong> {{ $user->ranks->name }}
```

```
                        </li>
                        <li class="list-group-item">
                            <strong>Location:</strong> {{ $user->locations->name }}
                        </li>
                        <li class="list-group-item">
                            <strong>Unit:</strong> {{ $user->units->name }}
                        </li>
                        <li class="list-group-item">
                            <strong>Created At:</strong>
                            {{ $user->created_at ? $user->created_at->format('d-m-Y H:i') :
'N/A' }}
                        </li>
                        <li class="list-group-item">
                            <strong>Last Updated:</strong>
                            {{ $user->updated_at ? $user->updated_at->format('d-m-Y H:i') :
'N/A' }}
                        </li>

                    </ul>

                </div>
            </div>
        </div>
    </div>
</div>
```

7.Yajra laravel datatables(https://yajrabox.com/docs/laravel-datatables/12.0)

- `composer require yajra/laravel-datatables-oracle:"^12.0"`
- `composer require yajra/laravel-datatables:"^12.0"`
- `php artisan vendor:publish --tag=datatables`
- (https://yajrabox.com/docs/laravel-datatables/12.0/quick-starter)

`npm i laravel-datatables-vite --save-dev`

`resources/js/app.js`

```
import 'laravel-datatables-vite';
```

`resources/sass/app.scss`

```
// DataTables
@import 'bootstrap-icons/font/bootstrap-icons.css';
@import "datatables.net-bs5/css/dataTables.bootstrap5.min.css";
@import "datatables.net-buttons-bs5/css/buttons.bootstrap5.min.css";
@import 'datatables.net-select-bs5/css/select.bootstrap5.css';
```

`npm run dev`

`php artisan datatables:make Users`

```
<?php

namespace App\DataTables;
```

```php
use App\Models\User;
use Illuminate\Database\Eloquent\Builder as QueryBuilder;
use Yajra\DataTables\EloquentDataTable;
use Yajra\DataTables\Html\Builder as HtmlBuilder;
use Yajra\DataTables\Html\Button;
use Yajra\DataTables\Html\Column;
use Yajra\DataTables\Html\Editor\Editor;
use Yajra\DataTables\Html\Editor\Fields;
use Yajra\DataTables\Services\DataTable;

class UsersDataTable extends DataTable
{
    /**
     * Build the DataTable class.
     *
     * @param QueryBuilder<User> $query Results from query() method.
     */
    public function dataTable(QueryBuilder $query): EloquentDataTable
    {
        return (new EloquentDataTable($query))
        ->addIndexColumn()
        ->addColumn('active', function ($usersdatatable) {
            switch ($usersdatatable->active) {
                case 0:
                    return '<span class="badge badge-danger">In-Active</span>';
                    break;
                case 1:
                    return '<span class="badge badge-success">Active</span>';
                    break;
            }
        })
        ->addColumn('role.name', function ($user) {
            return '<span class="badge badge-primary">' . $user->role->name. '</span>';
        })
        ->addColumn('action', function ($usersdatatable) {
            $btn = '<a href="'.route('users.edit',$usersdatatable->id).'" class="btn btn-xs
btn-warning" data-toggle="tooltip" title="Edit User" ><i class="fa fa-pen"></i></a> ';
            $btn .= '<form  action="' . route('users.destroy', $usersdatatable->id) . '"
method="POST" class="d-inline" onsubmit="return confirmDelete()" >
                            ' . csrf_field() . '
                                ' . method_field("DELETE") . '
                            <button type="submit"  class="btn bg-danger btn-xs  dark:bg-gray-
800 dark:hover:bg-gray-700 dark:focus:ring-gray-700" onclick="return confirm(\'Do you need to
delete this\');" data-toggle="tooltip" title="Delete">
                            <i class="fa fa-trash-alt"></i></button>
                            </form> </div>';
            $btn .= '<a href="'.route('users.show', $usersdatatable->id).'" class="btn btn-xs
btn-info" data-toggle="tooltip" title="View User" ><i class="fa fa-eye"></i></a> ';
            return $btn;
        })
        ->rawColumns(['active', 'action','role.name']);
    }

    /**
     * Get the query source of dataTable.
     *
     * @return QueryBuilder<User>
     */
    public function query(User $model): QueryBuilder
    {
```

```
        return $model->newQuery()->with([
             'role',
             'ranks',
            'locations',
            'units'
        ]);
    }
    /**
     * Optional method if you want to use the html builder.
     */
    public function html(): HtmlBuilder
    {
        return $this->builder()
                    ->setTableId('users-table')
                    ->columns($this->getColumns())
                    ->minifiedAjax()
                    ->orderBy(1)
                    ->selectStyleSingle()
                    ->buttons([
                        Button::make('excel'),
                        Button::make('csv'),
                        Button::make('pdf'),
                        Button::make('print'),
                        Button::make('reset'),
                        Button::make('reload')
                    ]);
    }

    /**
     * Get the dataTable columns definition.
     */
    public function getColumns(): array
    {
        return [
            // Column::make('DT_RowIndex')->title('#')->searchable(false)->orderable(false)-
>width(5),
            Column::make('id'),
            Column::make('service_no')->title('Service No')->data('service_no')-
>searchable(true),
            Column::make('name')->title('Name')->data('name')->searchable(true),
            Column::make('email')->title('Email')->data('email')->searchable(true),
            Column::make('active')->title('Active')->data('active')->searchable(true),
            Column::make('role.name')->title('Role')->data('role.name')->searchable(true),
            Column::make('ranks.name')->title('Rank')->data('ranks.name')->searchable(true),
            Column::make('locations.name')->title('Location')->data('locations.name')-
>searchable(true),
            Column::make('units.name')->title('Unit')->data('units.name')->searchable(true),
            Column::computed('action')
                  ->exportable(false)
                  ->printable(false)
                  ->width(80)
                  ->addClass('text-center'),
        ];
    }

    /**
     * Get the filename for export.
     */
    protected function filename(): string
    {
```

```
        return 'Users_' . date('YmdHis');
    }
}
```

8.Tricks

- How do I confirm delete?

In index.php

```
@if (session('status'))
                <div class="alert alert-success">{{ session('status') }}</div>
            @endif
```

```
 <a href="{{ route('users.edit', $user->id) }}" class="btn btn-xs btn-warning" data-
toggle="tooltip" title="Edit User" ><i class="fa fa-pen"></i></a>
                                    <form action="{{ route('users.destroy', $user->id) }}"
method="POST" style="display:inline;" onsubmit="return confirmDelete()" >
                                        @csrf
                                        @method('DELETE')
                                        <button type="submit"   class="btn btn-xs btn-danger"
data-toggle="tooltip"  title="Delete User" ><i class="fa fa-trash"></i></button>
                                    </form>
```

```
@endsection
    {{-- <script>
        function confirmDelete(){
            return confirm('Are You sure you want to delete this record? This action be
undone.');

        }

    </script> --}}
```

- How do you get back in after jumping out of the table?

In index.php

```
 <div class="table-responsive">
            {{ $dataTable->table() }}

        </div>
```

- How to install softdelete?
  - Create migration
  - Php artisan make:migration add_deleted_at_to_users_table
  - In user model

```
<?php

namespace App\Models;
```

```php
// use Illuminate\Contracts\Auth\MustVerifyEmail;
use Illuminate\Database\Eloquent\Factories\HasFactory;
use Illuminate\Foundation\Auth\User as Authenticatable;
use Illuminate\Notifications\Notifiable;
use Spatie\Permission\Traits\HasRoles;
use Illuminate\Database\Eloquent\SoftDeletes;

class User extends Authenticatable
{
    /** @use HasFactory<\Database\Factories\UserFactory> */
    use HasFactory, Notifiable ,HasRoles, SoftDeletes;
```

- In add_deleted_at_to users_table.php

```php
public function up(): void
    {
        Schema::table('users', function (Blueprint $table) {
            $table->softDeletes();
        });
    }
```

- How to make a connection?
  In user.php

```php
<?php

namespace App\Models;

// use Illuminate\Contracts\Auth\MustVerifyEmail;
use Illuminate\Database\Eloquent\Factories\HasFactory;
use Illuminate\Foundation\Auth\User as Authenticatable;
use Illuminate\Notifications\Notifiable;
use Spatie\Permission\Traits\HasRoles;
use Illuminate\Database\Eloquent\SoftDeletes;

class User extends Authenticatable
{
    /** @use HasFactory<\Database\Factories\UserFactory> */
    use HasFactory, Notifiable ,HasRoles, SoftDeletes;

    /**
     * The attributes that are mass assignable.
     *
     * @var list<string>
     */
    protected $fillable = [
            'id',
            'service_no',
            'name',
            'email',
            'email_verified_at',
            'password',
            'remember_token',
            'active_date_time',
            'active',
```

```
            'created_at',
            'updated_at',
            'location_id',
            'role_id',
            'rank_id',
            'unit_id',
    ];
    /**
     * The attributes that should be hidden for serialization.
     *
     * @var list<string>
     */
    protected $hidden = [
        'password',
        'remember_token',
    ];
    /**
     * Get the attributes that should be cast.
     *
     * @return array<string, string>
     */
    protected function casts(): array
    {
        return [
            'email_verified_at' => 'datetime',
            'password' => 'hashed',
        ];
    }
    public function ranks()
    {
        return $this->belongsTo(Rank::class, 'rank_id');
    }
    public function locations()
    {
        return $this->belongsTo(Location::class, 'location_id');
    }
    public function units()
    {
        return $this->belongsTo(Unit::class, 'unit_id');
    }
    public function role()
    {
        return $this->belongsTo(Role::class, 'role_id');
    }
}
```

## In controller

```
<?php

namespace App\Http\Controllers;
use App\Models\User;
use App\Models\Location;  // Import Location Model
use App\Models\Role;
use App\Models\Rank;
use App\Models\Unit;
use App\DataTables\UsersDataTable;
use Illuminate\Support\Facades\Hash;
```

```php
use Illuminate\Http\Request;

class UserController extends Controller
{
    public function __construct()
    {
        $this->middleware('permission:view_users')->only('index', 'show');
        $this->middleware('permission:create_users')->only('create', 'store');
        $this->middleware('permission:edit_users')->only('edit', 'update');
        $this->middleware('permission:delete_users')->only('destroy');

    }
    public function index(UsersDataTable $dataTable)
    {
        return $dataTable->render('users.index');
    }
    public function create()
    {
        $location = Location::all();
        $role = Role::all();
        $rank = Rank::all();
        $unit = Unit::all();
        return view('users.create', compact('location','role','rank','unit'));
    }
    public function store(Request $request)
    {
        $validated = $request->validate([
        'service_no' => 'required|string|max:255',
        'name' => 'required|string|max:255',
        'email' => 'required|email|unique:users,email',
        'password' => 'required|string|min:8',
         'location_id' => ['required', 'numeric'],
         'rank_id' => ['required', 'numeric'],
         'unit_id' => ['required', 'numeric'],
         'password' => ['required', 'string', 'min:8'], // Optional field
          'role_id' => ['required'], // Validate role input
        ]);

        $user = User::create($validated);
        $user->password = Hash::make($request->password);
        $user->active = 1;
        $user->save();

       $role = Role::findOrFail($request->input('role_id'));

        // Sync the roles for the user (in case role is updated)
        $user->syncRoles([$role->name]);
        $user->assignRole($role->name);

        return redirect()->route('users.index');
    }

    public function show(User $user)
    {
        return view('users.show', compact('user'));
    }
    public function edit(User $user)
    {
        $location = Location::all();
        $role = Role::all();
```

```
        $rank = Rank::all();
        $unit = Unit::all();
        return view('users.edit', compact('user','location','role','rank','unit'));
    }
    public function update(Request $request, User $user)
    {
        $user_detail = $request->validate([
            'service_no' => ['required', 'string'],
            'name' => 'required|string|max:255',
            'email' => 'required|string|max:255',
            'location_id' => ['required', 'numeric'],
            'rank_id' => ['required', 'numeric'],
            'unit_id' => ['required', 'numeric'],
            'password' => 'required|string|min:8', // Optional field
            'role_id' => ['required'], // Validate role input
        ]);
        $user->update($user_detail);
        $role = Role::findOrFail($request->input('role_id'));
        // Sync the roles for the user (in case role is updated)
        $user->syncRoles([$role->name]);
        $user->assignRole($role->name);

        $user->syncPermissions($request->input('permissions'));

        // $user->update($request->all());
        return redirect()->route('users.index');
    }
    public function destroy(User $user)
    {
        $user->delete();
        return redirect()->route('users.index');
    }
}
```

## In userDataTable

```
 public function query(User $model): QueryBuilder
    {
        return $model->newQuery()->with([
             'role',
             'ranks',
            'locations',
            'units'
        ]);
    }
```

## In unit.php

```
<?php

namespace App\Models;

use Illuminate\Database\Eloquent\Model;
use Illuminate\Database\Eloquent\SoftDeletes;
class Unit extends Model
{
    use SoftDeletes;
    protected $fillable = [
        'name',
            ];
    public function user()
    {
```

```
        return $this->hasmany(user::class);
    }
}
```

## In user controller

```
<?php

namespace App\Http\Controllers;
use App\Models\User;
use App\Models\Location;  // Import Location Model
use App\Models\Role;
use App\Models\Rank;
use App\Models\Unit;
use App\DataTables\UsersDataTable;
use Illuminate\Support\Facades\Hash;



use Illuminate\Http\Request;

class UserController extends Controller
{
    public function __construct()
    {
        $this->middleware('permission:view_users')->only('index', 'show');
        $this->middleware('permission:create_users')->only('create', 'store');
        $this->middleware('permission:edit_users')->only('edit', 'update');
        $this->middleware('permission:delete_users')->only('destroy');

    }
    public function index(UsersDataTable $dataTable)
    {
        return $dataTable->render('users.index');
    }

    public function create()
    {
        $location = Location::all();
        $role = Role::all();
        $rank = Rank::all();
        $unit = Unit::all();
        return view('users.create', compact('location','role','rank','unit'));
     }
    public function store(Request $request)
    {

        $validated = $request->validate([
        'service_no' => 'required|string|max:255',
        'name' => 'required|string|max:255',
        'email' => 'required|email|unique:users,email',
        'password' => 'required|string|min:8',
        'location_id' => ['required', 'numeric'],
        'rank_id' => ['required', 'numeric'],
        'unit_id' => ['required', 'numeric'],
         'password' => ['required', 'string', 'min:8'], // Optional field
         'role_id' => ['required'], // Validate role input
        ]);

        $user = User::create($validated);
        $user->password = Hash::make($request->password);
```

```
        $user->active = 1;
        $user->save();

        $user->syncRoles($request->input('role_id'));
        $user->assignRole($request->input('role_id'));
        return redirect()->route('users.index');
    }
    public function show(User $user)
    {
        return view('users.show', compact('user'));
    }
    public function edit(User $user)
    {
        $location = Location::all();
        $role = Role::all();
        $rank = Rank::all();
        $unit = Unit::all();
        return view('users.edit', compact('user','location','role','rank','unit'));
    }
    public function update(Request $request, User $user)
    {
        $user_detail = $request->validate([
            'service_no' => ['required', 'string'],
            'name' => 'required|string|max:255',
            'email' => 'required|string|max:255',
            'location_id' => ['required', 'numeric'],
            'rank_id' => ['required', 'numeric'],
            'unit_id' => ['required', 'numeric'],
            'password' => 'required|string|min:8', // Optional field
            'role_id' => ['required'], // Validate role input
        ]);
        $user->update($user_detail);
        $role = Role::findOrFail($request->input('role_id'));
        // Sync the roles for the user (in case role is updated)
        $user->syncRoles([$role->name]);
        $user->assignRole($role->name);

        $user->syncPermissions($request->input('permissions'));

        // $user->update($request->all());
        return redirect()->route('users.index');
    }
    public function destroy(User $user)
    {
        $user->delete();
        return redirect()->route('users.index');
    }
}
```

- How to create crud?

crud

ProductController

<?php

```
namespace App\Http\Controllers;

use App\Models\Product;
use Illuminate\Http\Request;

class ProductController extends Controller
{
    public function index()
    {
        $products = Product::all();
        return view('products.index', compact('products'));
    }

    public function create()
    {
        return view('products.create');
    }

    public function store(Request $request)
    {
        Product::create($request->all());
        return redirect()->route('products.index');
    }

    public function show(Product $product)
{
    return view('products.show', compact('product'));
}

    public function edit(Product $product)
```

```
    {
        return view('products.edit', compact('product'));
    }

    public function update(Request $request, Product $product)
    {
        $product->update($request->all());
        return redirect()->route('products.index');
    }

    public function destroy(Product $product)
    {
        $product->delete();
        return redirect()->route('products.index');
    }
}
```

web.php

```
use App\Http\Controllers\ProductController;
Route::resource('products', ProductController::class);
```

migration

```
<?php

use Illuminate\Database\Migrations\Migration;
use Illuminate\Database\Schema\Blueprint;
use Illuminate\Support\Facades\Schema;

return new class extends Migration
{
```

```
    /**
     * Run the migrations.
     */
    public function up(): void
    {
        Schema::create('products', function (Blueprint $table) {
            $table->id();
            $table->string('name');
            $table->string('category');
            $table->decimal('price', 8, 2);
            $table->integer('quantity');
            $table->timestamps();
        });
    }

    /**
     * Reverse the migrations.
     */
    public function down(): void
    {
        Schema::dropIfExists('products');
    }
};
```

model

```
<?php

namespace App\Models;
use Illuminate\Database\Eloquent\SoftDeletes;
```

```
use Illuminate\Database\Eloquent\Model;

class Product extends Model

{
    use SoftDeletes;

    protected $fillable = [
        'name',
         'category',
        'price',
        'quantity',
        // Add any other fields you need
    ];
}
```

add_deleted_at row migration

```
<?php

use Illuminate\Database\Migrations\Migration;
use Illuminate\Database\Schema\Blueprint;
use Illuminate\Support\Facades\Schema;

return new class extends Migration
{
    /**
     * Run the migrations.
```

```
     */
    public function up(): void
    {
        Schema::table('products', function (Blueprint $table) {
            $table->softDeletes();
        });
    }

    /**
     * Reverse the migrations.
     */
    public function down(): void
    {
        Schema::table('products', function (Blueprint $table) {
            //
        });
    }
};
```

index.blade.php

```
@extends('adminlte::page')

@section('content')
<div class="container mt-5">
    <div class="row justify-content-center">
        <div class="col-md-10">

            @if (session('success'))
                <div class="alert alert-success">{{ session('success')
}}</div>
```

```
        @endif
{{--
        @if (session('status'))
            <div class="alert alert-success">{{ session('status')
}}</div>
        @endif --}}
        <div class="card mt-3">
            <div class="card-header">{{ __('Products') }}</div>

            <div class="card-body">
                <a href="{{ route('products.create') }}" class="btn
btn-primary float-end">Add New Product</a>

                <table class="table table-bordered table-striped">
                    <thead>
                        <tr>
                            <th>Name</th>
                            <th>Category</th>
                            <th>Price</th>
                            <th>Quantity</th>
                            <th>Actions</th>
                        </tr>
                    </thead>
                    <tbody>
                        @foreach ($products as $product)
                        <tr>
                            <td>{{ $product->name }}</td>
                            <td>{{ $product->category }}</td>
                            <td>{{ $product->price }}</td>
                            <td>{{ $product->quantity }}</td>
```

```
                        <td>
                            <a href="{{ route('products.edit', $product->id) }}" class="btn btn-xs btn-warning" data-toggle="tooltip" title="Edit User" ><i class="fa fa-pen"></i></a>
                            <form action="{{ route('products.destroy', $product->id) }}" method="POST" style="display:inline;" onsubmit="return confirmDelete()" >
                                @csrf
                                @method('DELETE')
                                <button type="submit"   class="btn btn-xs btn-danger" data-toggle="tooltip"  title="Delete User" ><i class="fa fa-trash"></i></button>
                            </form>
                            <a href="{{ route('products.show', $product->id) }}" class="btn btn-xs btn-info" data-toggle="tooltip" title="Edit User" ><i class="fa fa-eye"></i></a>
                        </td>
                    </tr>
                @endforeach
            </tbody>
        </table>

        </div>
    </div>
</div>
@endsection
<script>
    function confirmDelete(){
```

```
        return confirm('Are You sure you want to delete this
record? This action be undone.');

    }

</script>
```

create.blade.php

```
@extends('adminlte::page')
@section('content')
<div class="container mt-5">
    <div class="row justify-content-center">
        <div class="col-md-7">

            @if (session('status'))
                <div class="alert alert-success">{{ session('status')
}}</div>
            @endif
            <div class="card">
                <div class="card-header">{{ __(' Add Product')
}}</div>
                <div class="card-body">
                    <form action="{{ route('products.store') }}"
method="POST">
                        @csrf
                        <div class="mb-3">
                            <label for="">Name:</label>
                            <input type="text" name="name" required
class="form-control"/>
                        </div>
```

```
                    <div class="mb-3">
                        <label for="">Category: </label>
                        <input type="text" name="category" required
class="form-control"/>
                    </div>
                    <div class="mb-3">
                        <label for="">Price: </label>
                        <input type="number" name="price" required
class="form-control"/>
                    </div>
                     <div class="mb-3">
                        <label for="">Quantity: </label>
                        <input type="number" name="quantity"
required class="form-control"/>
                    </div>
                     <div class="mb-3">
                        <button type="submit" class="btn btn-
primary">Save</button>
                    </div>

                </form>
            </div>
        </div>
    </div>
</div>
```

edit.blade.php

```
@extends('adminlte::page')

@section('content')
```

```
<div class="container mt-5">
    <div class="row justify-content-center">
        <div class="col-md-7">

            @if (session('status'))
                <div class="alert alert-success">{{ session('status')
}}</div>
            @endif
            <div class="card">
                <div class="card-header">{{ __(' Edit Product')
}}</div>
                <div class="card-body">
                    <form action="{{ route('products.update', $product-
>id) }}" method="POST">
                        @csrf
                        @method('PUT')
                        <div class="mb-3">
                            <label for="">Name:</label>
                            <input type="text" name="name" id="name"
class="form-control" value="{{ old('name', $product->name) }}"
required>
                        </div>
                        <div class="mb-3">
                            <label for="">Category: </label>
                            <input type="text" name="category" id="name"
class="form-control" value="{{ old('name', $product->name) }}"
required>
                        </div>
                        <div class="mb-3">
                            <label for="">Price: </label>
```

```
                    <input type="number" name="price" id="price"
class="form-control" value="{{ old('price', $product->price) }}"
step="0.01" required>

                </div>

                <div class="mb-3">
                    <label for="">Quantity: </label>
                    <input type="number" name="quantity"
id="quantity" class="form-control" value="{{ old('quantity',
$product->quantity) }}" required>
                </div>
                    <div class="mb-3">
                    <button type="submit" class="btn btn-
primary">Update</button>
                </div>

            </form>
        </div>
    </div>
</div>
</div>
```
show.blade.php
```
@extends('adminlte::page')

@section('content')
<div class="container mt-5">
    <div class="row justify-content-center">
        <div class="col-md-7">
```

```
        @if (session('status'))
            <div class="alert alert-success">{{ session('status')
}}</div>
        @endif
        <div class="card">
            <div class="card-header"><strong>{{ $product->name
}}</strong>
            </div></div>
            <div class="card-body">
                <ul class="list-group">
                    <li class="list-group-item">
                        <strong>Category:</strong> {{ $product-
>Category }}
                    </li>
                    <li class="list-group-item">
                        <strong>Price:</strong> ${{
number_format($product->price, 2) }}
                    </li>
                    <li class="list-group-item">
                        <strong>Quantity:</strong> {{ $product-
>quantity }}
                    </li>
                    <li class="list-group-item">
                        <strong>Created At:</strong> {{ $product-
>created_at->format('d-m-Y H:i') }}
                    </li>
                    <li class="list-group-item">
                        <strong>Last Updated:</strong> {{ $product-
>updated_at->format('d-m-Y H:i') }}
```

```
                    </li>
                </ul>
            </div>
        </div>
    </div>
</div>
```

- How to bycript password?
    - Php artisan tinker
      >Hash::make('yourPasswordHere');

❖ Sample Project
https://github.com/Samadhi-perera/ems.git

Samadhi